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CSA1717 – ARTIFICIAL INTELLIGENCE

CO1 – LIST LAB PROGRAMS PYTHON

EXP 5: Missionaries and Cannibals Problem

PROGRAM:

practical lab 5.py - C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 5.py (3.14.5)

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```
from collections import deque

start = (3, 3, 0)
goal = (0, 0, 1)

moves = [(1, 0), (2, 0), (0, 1),
         (0, 2), (1, 1)]

def valid(m, c):
    if not (0 <= m <= 3 and 0 <= c <= 3):
        return False

    if m > 0 and m < c:
        return False

    mr, cr = 3 - m, 3 - c

    if mr > 0 and mr < cr:
        return False

    return True

queue = deque([(start, [start])])
visited = {start}
solution = None

while queue:
    state, path = queue.popleft()

    if state == goal:
        solution = path
        break

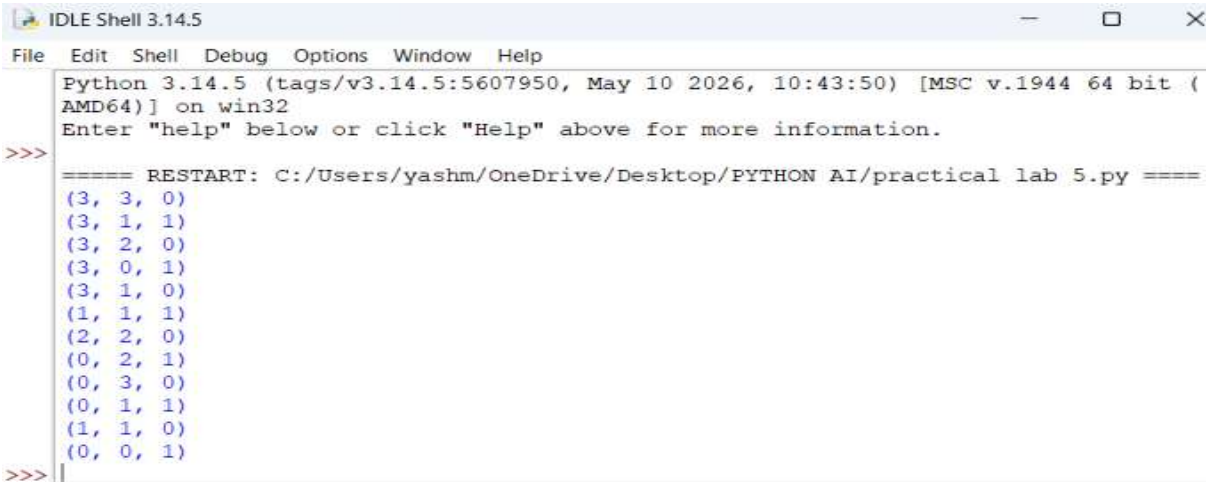
    m, c, boat = state

    for dm, dc in moves:
        if boat == 0:
            new_state = (m - dm, c - dc, 1)
        else:
            new_state = (m + dm, c + dc, 0)

        if (valid(new_state[0], new_state[1])
                and new_state not in visited):
            visited.add(new_state)
            queue.append((new_state, path + [new_state]))

if solution:
    for state in solution:
        print(state)
else:
    print("No solution found")
```

OUTPUT:


IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 5.py ====
(3, 3, 0)
(3, 1, 1)
(3, 2, 0)
(3, 0, 1)
(3, 1, 0)
(1, 1, 1)
(2, 2, 0)
(0, 2, 1)
(0, 3, 0)
(0, 1, 1)
(1, 1, 0)
(0, 0, 1)
>>>